GARBAGE SEGERIGATION AND BIN LEVEL DETECTION

A PROJECT REPORT

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BONAFIDE CERTIFICATE

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ABSTRACT

The public dustbins are overflowing and have become nobody's concern. Due to the lack of responsibility of the corporation people, the overflowing garbage wastes have created unhygienic surroundings and foul smell. So, to overcome this issue, a smart dustbin is designed. This smart dustbin is built on an Arduino Uno board and is interfaced with GSM, GPRS, and sensors. The sensors are used to check the threshold level of the dustbin. The threshold levels are already set. If the garbage hits the mentioned threshold level, continuous alerts are sent to the respective authority until the garbage is recovered and the externally fixed LED changes to red color. Once the garbage from the bin is cleared, the LED changes to green color. This alert system is triggered by the sensors to the GSM modem. A time limit is given to the respective authority, where if he/she fails the duty, the alert to the higher authority is sent. By this facility, the higher authority will be able to take action on the irresponsible workers.

Features like maps are used to locate the dustbins which make it easy for the authority to reach the location. Connectivity among the dustbins is given to establish communication among the bins and provides a smart system. Thus, the implementation of smart dustbins will create a hygienic society and will make the management of waste easy. The negligence of authorities and the public may be reduced. A clean and diseasefree environment can be created.

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LIST OF ABBREIVATION

1	GSM	Global System for Mobile Communications
2	GPRS	General Packet Radio Service
3	LED	Light Emitting Diode
4	LCD	Liquid Crystal Display
5	Wi-Fi	Wireless Fidelity
6	IDE	Integrated Drive Electronics
7	USB	Universal Serial Bus
8	IOT	Internet of Things
9	GUI	Graphical User Interface
10	IR	Infrared
11	PC	Personal Computer
12	SPI	Serial Peripheral Interface
13	PWM	Pulse Width Modulation
14	MHZ	Megahertz
15	RFID	Radio-Frequency Identification

CHAPTER I
INTRODUCTION

1.1. DOMAIN DESCRIPTION. Internet of things (IOT):

This dcscribcs the nctwork of physical objccts—"things"—that arc embedded with sensors, softwarc, and othcr tcchnologics for tlic purposc of connecting and cxchanging data with othcr dcviccs and systcins over thc lntcmct.

Things have evolved due to thc convcrgcncc of inultiplc tcclinologics, rcaltiinc analytics, machine learning, commodity sensors, and embedded systcins. Traditional fields of cinbcddcd systems, wirclcss scnsor networks, control systcins, automation (including home and building automation), and othcrs all contribute to cnabling thc Intcmct of things. In thc consumer inarkct, IOT technology is most synonymous with products pcrtaining to tlic conccpt of the " smart homc" , including devices and appliances (such as lighting fixturcs, thcnnostats, homc security systcins and caincras, and othcr homc appliances) that support onc or more common ccosystcins, and can bc controlled via devices associatcd with that ccosystcm, such as smart phoncs and smart spcakcrs. IOT can also be uscd in licalthcarc systems. Tlicrc arc a number of serious conccms about dangcrs in the growth of IOT, cspccially in the arcas of privacy and security, and conscqucntly industry and govcmnicntal niovcs to address these conccms have begun including thc dcvclopmcnt of intcmational standards[3].

The perfect network connectivity option would consume exwemely little power, have huge range, and would be able to transmit large amounts of data (high bandwidth). Unfortunately, this perfect connectivity doesn't exist. Each connectivity option represents a tradeoff between power consumption, range, and bandwidth. This allows us to create a conceptual framework that segments the various connectivity options into three major groups:

High Power Consumption, High Range, High Bandwidth - To wirclcssly scnd a lot of data over a grcat distance, it takes a lot of power. A great cxainplc of this is your smart phonc. Your phone can rcccivc and transmit largc amounts of data (c.g. vidco) ovcr grcat distanccs, but you nccd to charge it cvcry 1-2 days.

Conncctivity options in this group includc ccllular and satcllitc. Cellular is uscd when thc scnsor/device is within covcragc of cell towers. For scnsors/dcviccs that arc, say, on a ship in thc iniddlc of the ocean, satcllitc becomes ncccssary

Low Power Consumption, Low Range, High Bandwidth - To decrease power consumption and still send a lot of data, you have to decrease the range.

Connectivity options in this group include Wi-Fi, Bluetooth, and Ethernet. Ethernet is a hard-wired connection, so the range is short because it's only as far as the length ofthe cable. Wi-Fi and Bluetooth are both wireless connections with lower power consumption than cellular and satellite and with up to or greater than the bandwidth. However, as I'm sure you've experienced just walking

around your home, the range is limited.

Low Power Consumption, High Range, Low Bandwidth - To increase range while maintaining low power consumption, you have to decrease the amount of data that you're sending. Connectivity options in this group are called Low-Power Wide-Area Networks (LPWAN).

1 .2 ADVANTAGES OF IOT

- Easy ccess - igh now, you can easl y access the necessary Information m real-time; from (ålmost) whichever location you are at"All it takes iS smart device and internet connection: We use Google Maps to see where we are, instead of asking a person -in the•treet. Booking is•impler than information -is -easily accessibleceven fronvthe latestscientific research business analys is a click click away.

Speed - All this data pouring in enables us to complete numerous tasks with cnvying spccd. For example, loT makes automation casy. Smart offices automate repetitive tasks, thus allowing cmployces to invest their time and effort into something more challenging.

Adapting to New Standards - 1 hough IoT is ever-changing, its alterations are minimal compared to the rest of the high tech world. Without IoT, it would be hard for us to keep track of all the latest updates.

- Better Time Management - Overall, the IoT is an incredible time-saving We can search for the latest news on our phones during our daily commute, or visit a

blog about our lävorite pastime, purchase an item in an online shop, you name it.

1 .3 INTRODUCTION OF THE PROJECT.

Thc society, in the urgc of transforming a city into smart city, has to work on various sectors. Thc main sector is thc inanagcincnt of wastc. So the smart dustbin conccpt has bccn cvolvcd. This projcct involvcs the hardwarc componcnts likc Arduino Uno, Scrvo motors, connccting wircs and battcrics. It also inherits thc conccpt of GSM for SMS and GPRS for Maps to locatc thc dustbin.

Thc Ultrasonic sensors dctccts and cvaluatcs thc distancc of thc objcct. An Infiarc sensor cniits IR radiation in order to sense any actioinAn IR sensor dctccts thcmial radiation from Othcr obicctsnhc Arduino Unoinniicrocontrollcr board.

1 .4 PROBLEM STATEMENT

The objcctive is to dcsign and build an automatic opcn and close dustbin. Dustbins arc provided with a scnsor which helps in tracking thc lcvcl of thc garbagc in thc bins and a unique ID is givcn to cvcry dustbin in the city so that it is casy to idcntify which garbagc bin is tull. To provide casy acccssibility to all thc pcoplc, GUI is crcatcd which hclps pcoplc to know the location and status of thc dustbin.

1 .5 PROBLEM DESCRIPTION

With incrcasc of population, thc scenario of clcanlincss with respect to garbagc nianagcincnt is dcgrading tremendously. In citics thcrc arc many public placcs whcrc we sec that garbagc bins or dustbins arc placed but arc ovcrflowing. This creates unhygicnic condition in the nearby surrounding. Also crcatcs uglincss, scrious discascs, produces foul smell which dcgradcs tlic valuation of that arca. To avoid such situation, a project called ”Smart Dustbin" is dcvclopcd, which is a GSM based Garbagc and garbagc lcvcl in thc dustbin is indicated using LED lights, as wcll as thc GPRS to locatc thc dustbins.

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CHAPTER 2
LITERATURE SURVEY

2.1 STATE OF ART

Authors Twinkle Sinha, K. Mugcsh Kumar, P. Saisliara (2015)in thc paper about "SMART DUSTBIN, " - A single directional cylinder is suspcndcd ncxt to the lid of dustbin. Thc piston is frcc to move up and down vcrtically insidc thc dustbin to a ccrtain lcvcl. A platc is attached to thc cylinder for coinprcssing thc garbagc. Thc shapc of this plate dcpcnds upon thc shape of thc dustbin. Thc compressing platc consists of a side hole through which thc leaf switch is suspcndcd upsidc down. Thc lcvcl of lcaf switch is placed lowcr to the maximum lcvcl to which thc compressing platc can rcach down thus cvcn aftcr the switch gets pressed, garbagc can bc dunipcd in thc dustbin to a ccrtain cxtcnt.

In tlic paper "Smart dustbin for cconoinic growth", thc authors U. Nagaraju, Ritu Mishra, Chaitanya Kumar, Rajkuniar(2017) cnipliasizcd on thc conccpt of smart dustbin which accepts a SIM card, and operates over a subscription to a inobilc opcrator, just likc a mobile phone. A GSM modem can bc an cxtcmal dcvicc or a PC Card / PCMCIA Card. An cxtcrnal GSM modem is connected to a computer through a serial cablc or a USB cable. When a GSM modem is connected to a computer, this allows thc coinputcr to coinniunicatc ovcr thc mobilc network. While thcsc GSM modems arc most frcqucntly uscd to providc mobile for intcmct conncctivity, many of them can also be uscd for scnding and rccciving SMS and MMS incssagc. GSM Modcin sends and rcccivcs data through radio waves. In this projcct GSM 900 modem is uscd to scnd thc messages. It consists of a GSM/GPRS modem with standard communication intcrfaccs likc RS-232 (Scrial Port), USB, so that it can bc easily conncctcd to thc othcr dcviccs.
The power supply circuit is also built in thc inodulc that can bc turn ON by using a suitablc adaptor.
In thc papcr titled "Smart Dustbin- An Intclligcnt Approach to Fulfill Swatclili Bharat Mission", thc authors Priyain Parikh, Dr. Rupcsh Vasani, Akshar Rava (2017) had Interfaced RFID with Arduino UNO. Intcrfacing Scrial LCD with Arduino. Scndiiig AT commands to GSM inodulc. Dccision making with thc usc of ultrasonic sensor. Common DC power supply dcsigning. linplcincnting SPI communication in Arduino. Multiple Serial Port communication. Circuit Size Rcduction. Providing PWM signals to Servo Motor. Authors M.R. Sundarakuinar, S. Navccn, S. Arun kuinar, Kanial Kumar Ray (2017) in thc papcr "Smart dustbin garbagc monitoring system by cfficicnt arduino based" have uscd ultrasonic sensors used to incasurc thc garbagc height. Arduino is uscd to connect all the modules for data

full. Bluetooth module used to display the garbage height to the user mobile. Instead of normal battery the lithium battery and solar panel will be used because it can rechargeable. In the paper "Smart Waste Bin with

Real-Time Monitoring System" the authors
Norfadzlia Mohd Yusof, Mohd Faizal Zulkifli, Nor Yusma Amira Mohd Yusof, Azziana Afififie Azinan (2018) - As a guidelines for proposing the new framework for waste management system, several problems in waste management are considered:

1. Lack of information about the collecting time and area.
2. Lack of proper system to monitor the trucks and wash bins that have been collected in real time.
3. There is no estimation to the amount of solid waste inside the bin and the surrounding area due to the scattering of waste.
4. There is no quick response to urgent cases like truck accident, breakdown, longtime idling.
5. There is no quick way to response to client's complaints about uncollected waste.
6. There is no analysis of finding best route path of collecting waste.

The proposed confirmed able to provide solutions to all above problems. In this paper the framework is used to develop modules for problems 1, 2, 3 and 5. This framework consists of three segments namely the solar power system, smart waste bin and control station. Each part is facilitates with subsystems that will execute different tasks. All these segments and subsystems will be further described in the following section.

In the paper titled "IOT based garbage monitoring using Arduino" the authors Md. Aasim Anwar, Prateek Sarkar, Rajeshwar Dutta, Md. Sadik Mohammad Mollick (2018) — have designed structure of the system is developed before implementation of circuit. Used advanced microcontroller called Arduino (ATmega8). It is in-built with many components like analog to digital converter, clock of 16 MHz, shift registers. In this project we put the ultrasonic sensor on top of the garbage bin/ dump. The output of the ultrasonic sensor is processed by the Arduino and the output is then sent to the GSM module which sends a text message to the concerned person. We have a threshold value of 5cm. Which means that if the distance of the sensor from the top of the garbage is less than 5cm, the output will come with a message that the basket is full? Also, a buzzer will ring if output is less than 5cm. The DHTI I sensor will show the temperature and the humidity.

In the paper "Smart Dustbin" authors namely Dr. Baidyanath Ram Prajapati (2018) shave designed a "Smart Dustbin" which is a GSM enabled bin which automatically detects the garbage level and sends message to respective municipal authorities updating the status of the bin. To make our adopted village Kachhera Warsabad, G.B.
Nagar, UP clean and healthy using "Smac Dustbins". Authors Anilkumar C.S., Suhas G,

Sushma S (2019) in the paper "A Smart Dustbin using Mobile Application" discussed that there is no moving dustbin available in the market. It makes work simple for physically challenged people and aged who are unable to use in the dustbin and also provides a solution for Inappropriate placement of dustbin in the surrounding. A Smart Dustbin using Mobile Application. None of the system contains real time tracking status of the fullness of the bins. No dust bins are present that can detect the odour caused by the bins. In the paper "SMART DUSTBIN USING

ARDUINO," the authors Mamta Pandey,
Anamika Gowala, Mrinal Jyoti Goswami, ChininoySaikia and Dr. Dibyajyoti Bora. (2020) — Smart dustbin using arduino is an IOT based project. Here we are using arduino for code execution, for sensing we used ultrasonic sensor which will open lid and wait for few moment. It will bring drastic changes in terms of cleanliness with the help of technology. Everything is getting with smart technology for the betterment of human being. So this help in maintaining the environment clean with the help of technology. It is a sensor based dustbin so it would be easy to access/use for any age group. Our aim is also to make it cost effective so that many numbers of people can get the benefit from this. And it should be usable to anyone and helpful for them.

2.2 INFERENCE FROM LITERATURE

From all the above Literature reviews it is clearly understood that there are various ideas suggested for the Smart dustbin concept and various ideas suggested to deal with the garbage collected. But from the user's point of view, the perspective varies. This system helps users to take responsible decisions by allowing them to act as a responsible citizen for not taking the waste management for granted, since it is a serious challenge that every country all around the world is facing. This can be achieved by providing the users with a GUI that gives them clear information about the location of every dustbin in the area. Even if one dustbin is full they can easily access another nearby dustbin by easily viewing the status and location of each dustbin in the GUI.
Another important facility that this system provides is the LED light indicator system. This helps all the users to understand the capacity of the dustbin and act accordingly. People who are using the dustbins, either literate or illiterate can understand the capacity of the dustbin and take decisions accordingly.

CHAPTER 3 SYSTEM ANALYSIS

The existing project work is the implementation of Automatic Garbage Fill Alerting system

3.1 EXISTING SYSTEM using Ultrasonic sensor, Arduino Uno, Buzzer and Wi-Fi module. This system assures the cleaning of dustbins soon when the garbage level reaches its maximum. It will take power supply with the help of Piezoelectric Device. If the dustbin is not cleaned in specific time, then the record is sent to the Sweeper or higher authority who can take appropriate action against the concerned contractor. This system also helps to monitor the fake reports and hence can reduce the comiption in the overall management system[6]. This reduces the total number of trips of garbage collection vehicle and hence reduces the overall expenditure associated with the garbage collection. It ultimately helps to keep cleanliness in the society.
A few drawbacks of the existing system are as follows[4].

GPRS coordinates is not available to get the location of the dustbins.

GUI could be used to easily visualization.

Power constraints can be handled more efficiently.

Al technology can be used to collect garbage from the filled dustbins first.

3.2 PROPOSED SYSTEM

From the existing system, this project advances in features like maps and the automatic lock of the lid and remote access. Nowadays, maps play a major importance in various applications like finding a shortest route among various others, distance, to locate a vehicle like cabs etc,. In this project, the maps are used to locate the dustbins and helps the respective authority to get quick and easy to the location. The locking system is used to avoid overflow of trash and the spreading of foul smell around the area.

Even after the dustbin is full, people may try to dispose trash into dustbin. But the dustbin is designed in such a way that the lid is locked. This may confuse the person, literate or illiterate, that the dustbin is malfunctioning. So, to alert the person that the garbage is full, LED lights can be used. The LED lights has to be placed in the front, in a easily visible place. When the dustbin is full the LED light has to turn into red color. When the dustbin is not full the LED light has to turn into green color.

For the people in the control room to understand the status of the dustbin, a GUI is used. It can either be a website or an app. It should provide the garbage locations using maps and the status of the garbage. The application database also holds the details of the respective authorities and their details. Advantages of the proposed system are as follows.

GPRS coordinates is available to get the location of the dustbins. GUI

can be used for easy visualization.

Power constraints is handled more efficiently.

Al technology can be used to collect garbage from the filled dustbins first.

3.3 HARDWARE REQUIREMENTS

The most common set of requirements defined by any operating system or software application is the physical computer resources, also known as hardware. A hardware requirements list is often accompanied by a hardware compatibility list, especially in case of operating systems. The minimal hardware requirements are as follows,

- PROCESSOR : INTEL CORE 15
- RAM : 4GB
- PROCESSOR : 2.4 GHZ
- MAIN MEMORY : 4B RAM
- PROCESSING SPEED : 600 MHZ
- HARD DISK DRIVE: ITB
- KEYBOARD :104 KEYS • Arduino Uno 9.

- Ultrasonic sensor
- IR sensor

- Servo motor
- GSM module SIM800L
- SIM card 14. LED lights 15. GPRS module neo 6M. 16. Jumper wires (MTM/MTF/FTF) 17.
- 1k ohm resistors 18. 9V external batteries 19. l2V 1 AMP adaptor 20. Bread board.

3.4 SOFTWARE REQUIREMENTS

Software requirements deals with defining resource requirements and prerequisites that needs to be installed on a computer to provide functioning of an application. These requirements are need to be installed separately before the software is installed. The minimal software requirements are as follows,

1. IDE
2. ARDUINO IDE
3. OPERATING SYSTEM :WINDOWS 10

3.5 FEASIBILITY STUDY

The subject of the study was an examination of the possibilities for improving the existing waste management system, reducing costs, increasing efficiency and increasing profits. In addition to the economic motives, a significant driving factor was also the improvement of waste management, which would, in turn, increase the protection of the environment. By monitoring the availability of bins, it is possible to plan routes of the vehicles for discharging full bins without unnecessarily going to places of empty or half-empty bins, which would reduce fuel consumption and vehicle wearing (depreciation), and increase the effective work of workers and drivers. In addition, the company would have information on the possible disposal of recyclable waste, and could make the recasting of unprofitable and uneconomic placement courts. The aim of the study was to propose the introduction of the IOT concept in order to improve the waste management system.

CHAPTER 4

METHODOLOGIES

4.1 LANGUAGE DESCRIPTION

The Arduino Integrated Development Environment (IDE) is a cross-platform application (for Windows, mac OS, Linux) that is written in functions from C and C++. It is used to write and upload programs to Arduino compatible boards, but also, with the help of thirdparty cores, other vendor development boards. The source code for the IDE is released under the GNU General Public License, version 2. The Arduino IDE supports the languages C and C++ using special rules of code structuring. The Arduino IDE supplies a software library from the Wiring project, which provides many common input and output procedures. User-written code only requires two basic functions, for starting the sketch and the main program loop, that are compiled and linked with a program stub main () into an executable cyclic executive program with the GNU toolchain, also included with the IDE distribution. The Arduino IDE employs the program avrdude to convert the executable code into a text file in hexadecimal encoding that is loaded into the Arduino board by a loader program in the board's firmware. By default, avrdude is used as the uploading tool to flash the user code onto official Arduino boards. Arduino IDE is a derivative . Finally, Arduino provides a standard form factor that breaks out the functions of the microcontroller into a more accessible package. The Arduino is a microcontroller board based on the ATmega8. It has 14 digital -input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, al 6 MHz ceramic resonator, a USB connection, a power jack, an ICSP header, and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with a AC to-DC adapter or battery to get started . The Uno differs from all preceding boards in that it does not use the FTDI USBto-serial driver chip

42.1 Features ofBLYNK application • Similar API & UI for all supported hardware & devices

• Connection to the cloud using: • Wi-Fi • Bluetooth and BLE • Ethernet • USB (Serial) • GSM • Set of easy-to-use Widgets • Direct pin manipulation with no code writing • Easy to integrate and add new functionality using virtual pins • History data monitoring via Super-Chart widget • Device-to-Device communication using Bridge Widget • Sending emails, tweets, push notifications, etc. You can find example sketches covering basic Blynk

Features. They are included in the library. All the sketches are designed to be easily combined with each other.

of the Processing IDE, however as of version 2.0, the Processing IDE will be replaced with the Visual Studio

Code-based Eclipse Theia IDE framework.

4.2 ABOUT BLYNK APPLICATION

Blynk was designed for the Internet of Things. It can control hardware remotely, it can display sensor data, it can store data, visualize it and do many other cool things. There are three major components in the platform:

l. Blynk App - allows to you create amazing interfaces for your projects using various widgets we provide.

2. Blynk Server - responsible for all the communications between the smart phone and hardware. You can use our Blynk Cloud or run your private Blynk server locally. It's open-source, could easily handle thousands of devices and can even be launched on a Raspberry Pi. 3. Blynk Libraries - for all the popular hardware platforms - enable communication with the server and process all the incoming and outgoing commands.

Features of BLYNK application • Similar API & UI for all supported hardware & devices

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Features. They are included in the library. All the sketches are designed to be easily combined with each other.

4.3 IOT DEVICES USED

e 10T devices help in collecting very minute data from the surrounding environment. All
this collected data can have various degrees of complexities ranging from a simple
temperature monitoring sensor or a complex full video feed. A device can have multiple
sensors that can bundle together to do more than just sense things.

4 3.1 Arduino UNO

Arduino is an open-source platform used for building electronics projects. Arduino consists of both a physical programmable circuit board (often referred to as a microcontroller) and a piece of software, or IDE (Integrated Development Environment) that runs on your computer, used to write and upload computer code to the physical board[1].

• The Arduino platform has become quite popular with people just starting out with electronics, and for good reason. Unlike most previous programmable circuit boards, the Arduino does not need a separate piece of hardware (called a programmer) in order to load new code onto the board — you can simply use a

USB cable. Additionally, the Arduino IDE uses a simplified version of ++ making it easier to learn to program. Finally, Arduino provides a standard form factor that breaks out the functions of the micro-controller into a more accessible package. The Arduino is a microcontroller board based on the ATmega8. It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, al 6 MHz ceramic resonator, a USB connection, a power jack, an ICSP header, and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with a AC to-DC adapter or battery to get started . The Uno differs from all preceding boards in that it does not use the FTDI USB-to-serial driver chip. Instead, it features the Atmega l6U2 (Atmega8U2 up to version R2) programmed as a USB-to-serial converter .Revision 2 of the Uno board has a resistor pulling the 8U2HWB line to ground, making it easier to put into DFU mode.

Revision of the board has the following new features:

• Pin out: added SDA and SCL pins that are near to the AREF pin and two other new pins placed near to the RESET pin, the IOREF that allow the shields to adapt to the voltage IOT BASED GARBAGE MONITORING SYSTEM USING ARDUINO provided from the board. In future, shields will be compatible with both the board that uses the AVR, which operates with 5V and with the Arduino Due that operates with 3.3 V. The second one is a not connected pin that is reserved for future purposes.

"Uno" means one in Italian and is named to mark the upcoming release of Arduino 1.0. The Uno and version 1.0 will be the reference versions of Arduino, moving forward. The Uno is the latest in a series of USB Arduino boards, and the reference model for the Arduino platform.

Fig 4.1. Arduino Uno Board.

Parameters For Arduino UNO Description

Microcontrollcr	Arduino
Opcrating Voltagc	UNOSV
1 Input Voltage (rccomincndcd)	7- I2V
Input Voltagc (limits)	6-20V
1 Digital l/O Pins	14 6

DC Current per I/O Pin	40 inA
DC Current for 3.3V	50 mA

Pin Flash Memory SRAM	32KB(ATinega32S) of which 0.5KB used by boot loader 2 KB (ATnicga32S)
EEPRO	1 KB (ATincga328)
C1occcd k Sp	16MHz
Length	65.6 mm

Width Weight	53.4 mm 53gm

Tablc 4.1: Arduino UNO dcscription.

4 .3 .2 Ultrasonic sensor HC-SR04 is an ultrasonic scnsor which is uscd for nicasuring the distance bctwccn the top of the lid to thc top of thc garbagc.

PIN NUMBER	PIN NAME	DESCRIPTION
1. 2.	c Trigger	c cc Pin powers sensor, typically with +5V rigger pin is an Input pin. his pin has to be kept high for I Ous to initialize in c asurcnicnt by sending

		US wave.
3.	Echo	Echo pin is an Output pin. This pin goes high for a period of time which will be equal to the time taken for the US wave to return back to the sensor.
4.		This pin is connected to the Ground of the system.

Table 2: Pins and Functioning of Ultrasonic sensor The

following are the features of Ultrasonic sensor:

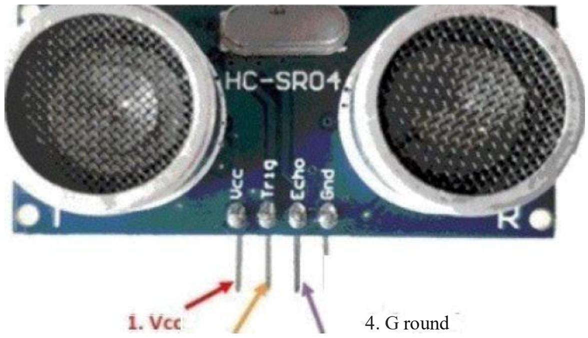
- Operating voltage: +5V
- Theoretical Measuring Distance: 2cm to 450cm
- Accuracy: 3mm
- Measuring angle covered: <15 ^{0}
- Operating Current: < l5ma
- Operating Frequency: 40Hz

The HC-SR04 Ultrasonic (US) sensor is a 4 pin module, whose pin names are Vcc, Trigger, Echo and Ground respectively. This sensor is a very popular sensor used in many applications where measuring distance or sensing objects are required. The module has two eyes like projects in the front which forms the Ultrasonic transmitter and Receiver. The sensor works with the simple high school formula that

Distance Speed x Time

The Ultrasonic transmitter transmits an ultrasonic wave, this wave travels in air and when it gets objected by any material it gets reflected back toward the sensor this reflected wave is observed by the Ultrasonic receiver module. Now, to calculate the distance using the above formulae, we should know the Speed and time. Since we are using the Ultrasonic wave we know the universal speed of US wave at room conditions which is 330m/s. The circuitry inbuilt on the module will calculate the time taken for the

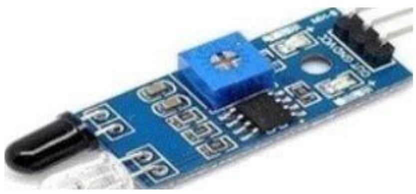
US wavc to comc back and turns on thc ccho pin high for that same particular amount of tiinc, this way we can also know the tiinc takcn. Now simply calculatc thc distance using a niicrocontrollcr or niicroproccssor[5].



2. Trigger 3. Eho

Fig 4.2: Ultrasonic scnsor

4.3.3 Infrarcd scnsor An infrared (IR) sensor is an clcctronic dcvicc that incasurcs and dctccts infrared radiation in its surrounding cnvironnicnt. Infrarcd radiation was accidcntally discovcrcd by an astronoincr named William Hcrchcl in 1500. Whilc incasuring thc tcmpcraturc of cach color ot light (scparatcd by a prism), hc noticcd that the tcinpcraturc just beyond thc rcd light was liighcst. IR is invisiblc to the human cyc, as its wavclcngtli is longer than that of visible light (though it is still on thc same clcctroniagnctic spcctrum). Anything that cinits hcat (cvcrything that has a tcnipcraturc above around fivc dcgrccs Kelvin) gives ofi infrarcd radiation. Tlierc arc two typcs of intrarcd scnsors: active and passive. Activc infrarcd sensors both cinit and dctcct infrared radiation. Activc IR sensors havc two parts: a light cinitting diodc (LED) and a rcccivcr. When an objcct comes closc to thc sensor, thc infrared light from thc LED rcflccts ofi of the objcct and is dctcctcd by tlic rcccivcr. Active IR scnsors act as proximity sensors, and they arc commonly uscd in obstaclc dctcction systcins (such as in robots).



4 3.4 Servo motor A servo motor is a motor whosc shaft turns to position something based olT a control signal. They arc typically uscd to stccr rcinotc control airplancs by adjusting thc wing flaps, flight position for droncs, controlling valves uscd in flow control or continuous drivc of wliccls for robots. Thcy can bc uscd to position or adjust almost anything you can think of. Thcy consist of a plastic housing which contains a DC motor, a control circuit and a few gears for torquc.

Servo motor control of thc shaft position conics from using a pulse width modulation signal (PWM) to turn the shaft clockwisc or countcr clockwisc, dcpcnding on the pulsc width of the signal. Typically, a pulse width of I ins will rotatc thc shaft clockwise and a 2 me pulsc will rotatc the shaft counter clockwise. To position the shaft $^{1}/2$ way, or in thc niiddlc, a 1.5 ins pulse typically works. You will nccd 20 nos bctwccn cach pulsc.

Fig 4.4: Servo motor

4 3.5 GSM niodulc SIMS()()L SIM800L is a ininiaturc ccllular inodulc which allows for

GPRS transmission, sending and rccciving SMS and making and receiving voicc calls. Low cost and small footprint and quad band frcqucncy suppor niakc this inodulc pcrtcct solution for any projcct that acquire long rangc conncctivity. Atlcr connccting power module boots up, scarchcs for ccllular network and login automatically. On board LED displays conncction state (no network covcragc - fast blinking, logged in - slow blinking).

This inodulc liavc two antcnnas included. First is made of wire (which solders directly to NET pin on PCB) - vcry useful in narrow placcs. Sccond - PCB antenna - with double sided tapc and attaclicd pigtail cablc with IPX conncctor. This onc havc bcttcr performance and allows to put your module insidc a nictal casc - as long the antenna is outsidc. Tlic following are tlic specifications of the GSM module. • Supply voltage: 3.8V - 4.2V

• Rccoinincndcd supply voltagc: 4V

GSM transmission (avg): 350inA

GSM transmission (pcck): 2000inA

- Module size: 25 x 23 non
- Intcrfacc: UART (max. 2.SV) and AT commands
- SIM card socket: micro SIM (bottom sidc)
- Supported frcqucncics: Quad Band (850 / 950 / 1500 / 1900 MHz)
- Antenna connector: IPX
- Status signaling: LED
- Working tcnipcraturc rangc: -40 do + 85 C

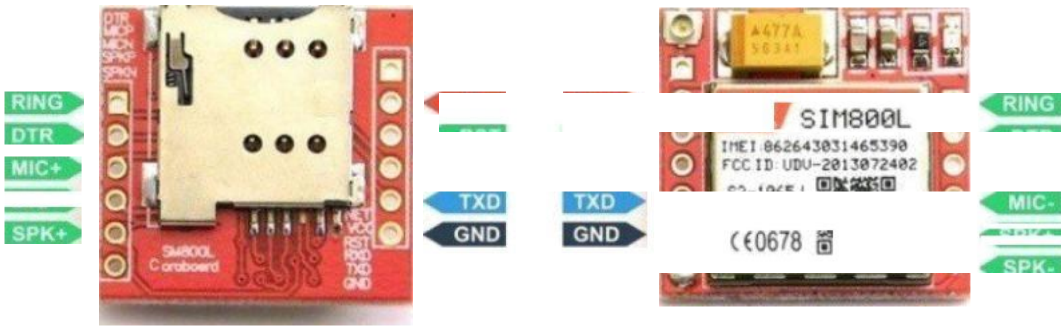


Fig 4.5: SIM8()()L module pin-out diagram.

- RING (not marked on PBC, first from top, square) - LOW state while receiving

call communication disabled). After setting it in LOW the module will wake up.

- DTR - sleep mode. Default in HIGH state (module in sleep mode, serial
- MICP, MICN - microphone (P + / N

- SPKP, SPKN - speaker (P + /N-)
- NET - antenna
- VCC - supplv voltage
- RESET - reset
- RXD - serial communication
- TXD - serial communication

GND - ground

.6 GPS module NE 0-6M [10]At the heart of the is a NEO-6M GPS chip from u-blox. The chip measures less than the size of a postage stamp but packs a surprising amount of features into its little frame.

NEO-6M GPS Chip

Fig 4.6: [IO]NEO - 6M GPS

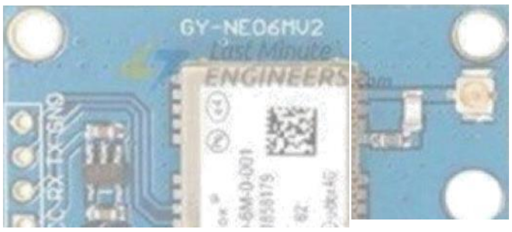
It can track up to 22 s ate lli tesGHIYR) channels and achieves the industry's highest level of sensitivity i.e. -161 dB tracking, while consuming onlv 45mA supply current.

[I O]UnIike other GPS modules, it can do up to 5 location updates a second with 2.5m Horizontal position accuracv The u-blox 6 positioning engine also boasts a Time-ToFirst-Fix (TTFF) of under I second. [10]One of the best features the chip provides is Power

Save Mode(PSM). It allows a reduction in svstem power consumption by selectively switching parts of the receiver ON and OFF. This dramatically reduces power consumption of the module to just 1 lmA making it suitable for power sensitive applications like GPS wristwatch.

[I O] The necessary data pins of NEO-6M GPS chip are broken out to a 0.1 " pitch headers. This includes pins required for communication with a microcontroller over UART. The module supports baud rate from 4800bps to 230400bps with default baud of 9600.

Receiver Tvpe 50 channels, GPS LIU 575.42Mhz)
Horizontal Position Accuracv 2.5m
Navigation Update Rate IHZ
(5Hz maximum) Capture Time Cool start:
27sHot start: Is Navigation Sensitivitv -161dBm
Communication Protocol NMEA, UBX Binary,
Scrial Baud Rate 4500-230400 (dcfault
Opcrating Tcinpcraturc -40 "C 85 "C 2.7V
Opcrating Voltagc 3.6V 45inA 5100
Operating Current TXD/RXD
lmpedanee



Position Fix LED Indicator

Fig 4.7: [10]Positionfix LED indicator.

[1 O] There is an LED on the NEO-6M GPS Module which indicates the status of Position Fix. It'll blink at various rates depending on what state it's in:

No Blinking — It's searching for satellites.

Blink every 1 s — Position Fix is found(The module can see enough satellites).

3.3V LDO Regulator

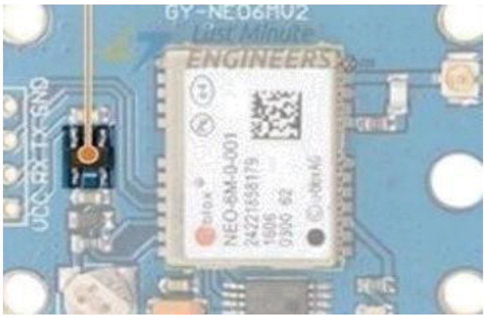


Fig 4.8: [10]3.3V LDO regulator

The operating voltage of the NEO-6M chip is from 2.7 to 3.6V. But the good news is that, the module comes with MIC5205 ultra-low dropout 3 V3 regulator from MICREL. The logic pins are also 5-volt tolerant, so we can easily connect it to an Arduino or any 5V logic microcontroller without using any logic level converter.

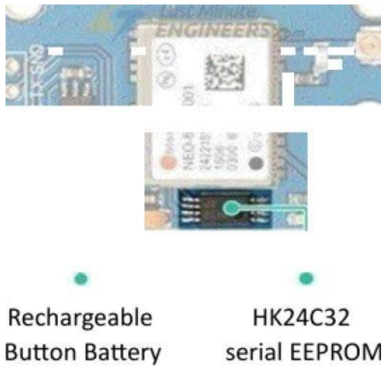


Fig 4.9: 0]Bu//eand EEPROM

[1 O] Thc inodulc is cquippcd with an HK24C32 two wirc scrial EEPROM. It is 4KB in size and conncctcd to the NEO-6M chip via 12C. Thc inodulc also contains a rccliargcablc button battery which acts as a super-capacitor. [1 O]An EEPROM togcthcr

with battery hclps retain thc battery backed RAM (BBR). Thc

BBR contains clock data, latest position data(GNSS orbit data) and niodulc contiguration.

But it's not incant for permanent data storage. [l O]As thc battcry rctains clock and last position, time to first fix (TTFF) significantly reduces

to Is. This allows much faster position locks.

[10] Without the battery the GPS always cold-start so thc initial GPS lock takes more tinic.Thc battery is automatically charged whcn powcr is applicd and maintains data for up to two wccks without powcr.

[1 O]An antenna is acquired to use tlic inodulc for any kind of communication. So, the inodulc coincs with a patch antenna having -161 dBni scnsitivity.

Fig 4.10: [IO]GPRS antcnna

You can snap-tit this antcnna to small U FL conncctor located on thc module.

U.FL connector

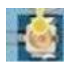


Fig 4.11: [IO]U.FL conncctor

Patch antenna is great for most projects. But if you want to achicvc inorc sensitivity or put your niodulc inside a nictal casc, you can also snap on any 3 V activc GPS antcnna via the U.FL conncctor. The NEO-I M GPS module has total 4 pins that intcrfacc it to thc outsidc world. The connections arc as follows:

Fig 4.12: [10]NEO-6M GPS niodulc pin-out

GND is the Ground Pin and nccds to be conncctcd to GND pin on the Arduino. TxD (Transmitter) pin is uscd for scrial communication. RxD (Rcccivcr) pin is uscd for scrial communication. VCC supplics powcr tor the niodulc. You can dircctly conncct it to thc 5 V pin on the Arduino.

4.4 ARDUINO IDE:

The Arduino Integrated Development Environment - or Arduino Software (IDE) - contains a text editor for writing code, a message area, a text console, a toolbar with buttons for common functions and a series of menus. It connects to the Arduino and Genuino hardware to upload programs and communicate with them.

4 4.1 Sketchbook

The Arduino Software (IDE) uses the concept of a sketchbook: a standard place to store your programs (or sketches). The sketches in your sketchbook can be opened from the File Sketchbook menu or from the Open button on the toolbar. The first time you run the Arduino software, it will automatically create a directory for your sketchbook. You can view or change the location of the sketchbook location from with the Preferences dialog.

4 4.2 Tabs, multiplefiles and compilation Allows you to manage sketches with more than one file (each of which appears in its own tab). These can be normal Arduino code files (no visible extension), C files (.c extension), C++ files (.cpp), or header files (.h).

4 4.3 Uploadmg

Before uploading your sketch, you need to select the correct items from the Tools Board and Tools Port menus. The boards are described below. On the Mac, the is probably something like /dev/tty.usbmodem24l (for an Uno or Mega2560 or Leonardo) or /dev/tty.usbserial-lBl (for a Dueinilanove or earlier USB board), or /dev/tty.USA l9QW lb 1P1.1 (for a serial board connected with a Key span USB- to-Serial adapter). On Windows, it's probably COMI or COM2 (for a serial board) or COM4,serial port COM5, COM 7, or higher (for a USB board) - to find out, you look for USB serial device in the ports section of the Windows Device Manager. On Linux, it should be /dev/ttyACMx , /dev/ttyUSBx or similar. Once you've selected the correct serial port and board, press the upload button in the toolbar or select the Upload item from the Sketch menu. Current Arduino boards will reset automatically and begin the upload. With older boards (pre-Diecimila) that lack auto-reset, you'll need to press the reset button on the board just before starting the upload. On most boards, you'll see the RX and TX LEDs blink as the sketch is uploaded. The Arduino Software (IDE) will display a message when the upload is complete, or show an error. When you upload a sketch, you're using the Arduino bootloader, a small program that has been loaded on to the microcontroller on your board. It allows you to upload code without using any additional hardware. The bootloader is active for a few seconds when the board resets; then it starts whichever sketch was most recently uploaded to the microcontroller. The boot loader will blink the on-board (pin 13) LED when it starts (i.e. when the board resets).

4.4.4 Libraries

Libraries provide extra functionality for use in sketches, e.g. working with hardware or manipulating data. To use a library in a sketch, select it from the Sketch Import Library menu. This will insert one or more #include statements at the top of the sketch and compile the library with your sketch. Because libraries are uploaded to the board with your sketch, they increase the amount of space it takes up. If a sketch no longer needs a library, simply delete its #include statements from the top of your code. There is a list of libraries in the reference. Some libraries are included with the Arduino software. Others can be downloaded from a variety of sources or through the Library Manager. Starting with version I .0.5 of the IDE, you do can import a library from a zip file and use it in an open sketch. See these instructions for installing a third-party library.

4.4.5 Third — party hardware

Support for third-party hardware can be added to the hardware directory of your sketchbook directory. Platforms installed there may include board definitions (which appear in the board menu), core libraries, boot loaders, and programmer definitions. To install, create the hardware directory, then unzip the third-party platform into its own sub-directory. (Don't use "arduino" as the sub-directory name or you'll override the built- in Arduino platform.) To uninstall, simply delete its directory.

4.4.6 Serial monitor

This displays serial sent from the Arduino or Genuino board over USB or serial connector.

To send data to the board, enter text and click on the "send" button or press enter. Choose the baud rate from the drop-down menu that matches the rate passed to Serial.begin in your sketch. Note that on Windows, Mac or Linux the board will reset (it will rerun your sketch) when you connect with the serial monitor. Please note that the Serial Monitor does not process control characters; if your sketch needs a complete management of the serial communication with control characters, you can use an external terminal program and connect it to the COM port assigned to your Arduino board.

4.5 UML DIAGRAMS

UML stands for Unified Modeling Language. UML is a standardized general-purpose modeling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group.

• The goal is for UML to become a common language for creating models of object oriented computer software. In its current form UML is comprised of two major components: a Meta-model and a notation. In the future, some form of method or process may also be added to; or associated with, UML.

• The Unified Modeling Language is a standard language for specifying, Visualization, Constructing and documenting the artifacts of software system, as well as for business modeling and other non-software systems.

• The UML represents a collection of best engineering practices that have proven successful in the modeling of large and complex systems.

• The UML is a very important part of developing objects oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

45.1 Goals The Primary goals in the design of the UML are as follows:

• Provide users a ready-to-use, expressive visual modeling Language so that they can develop and exchange meaningful models.

• Provide extensibility and specialization mechanisms to extend the core concepts.

Be independent of particular programming languages and development process.

• Provide a formal basis for understanding the modeling language. • Encourage the growth of 00 tools market.

• Support higher level development concepts such as collaborations, frameworks, patterns and components. • Integrate best practices.

4 5.2 System architecture diagram

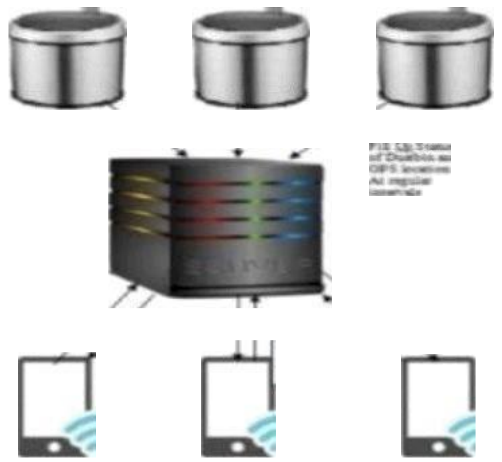
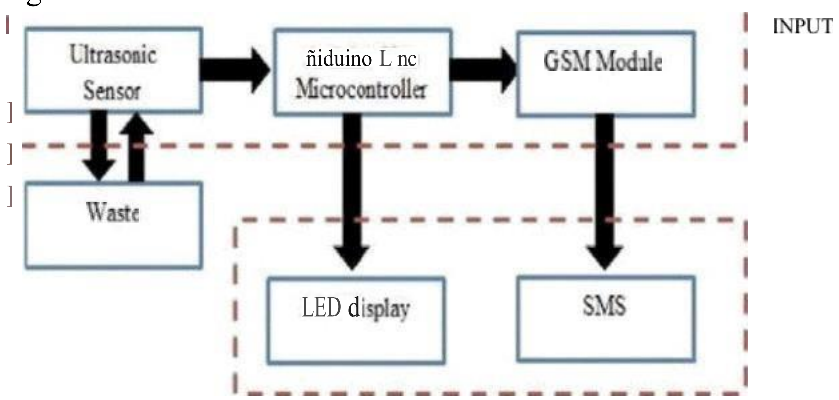


Fig 4.13. System Architecturc diagram **

4 5.3 Block diagram A block diagram is a diagram ot a systcin in which the principal parts or

functions arc rcprcscntcd by blocks conncctcd by lincs that show thc relationships of the blocks.
They arc heavily uscd in cnginccring in hardware dcsign, clcctronic dcsign, softwarc dcsign, and proccss flow diagrams.



OUTPUT

Fig 4.14: Block diagram

4 5.4 Sequence diagram A sequence diagram in Unificd Modcling Languagc (UML) is a kind ot intcraction diagram that shows how proccsscs operate with onc another and in what order. It is a construct of a Message Scqucncc Chart. Scqucncc diagrams arc sonictinics callcd cvcnt diagrams, event scenarios, and tinting diagrams.

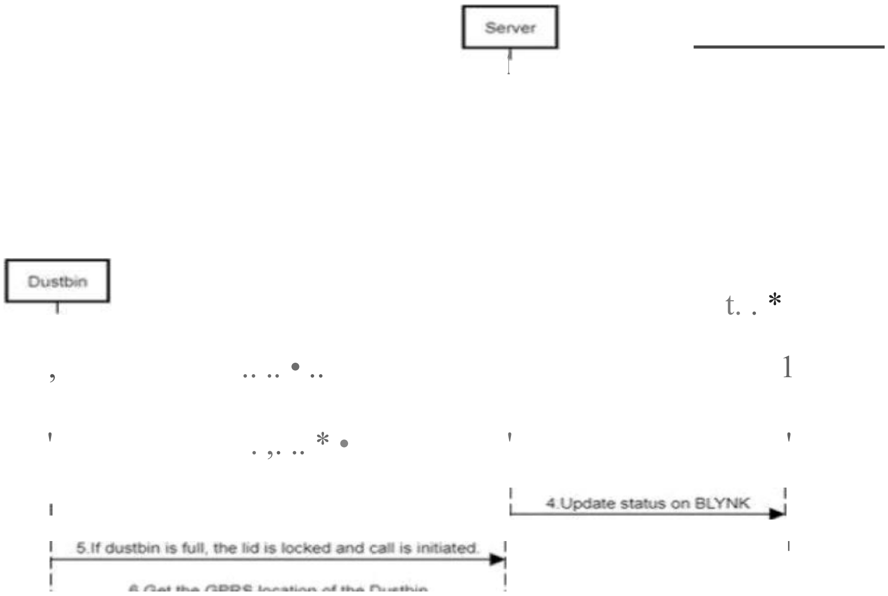
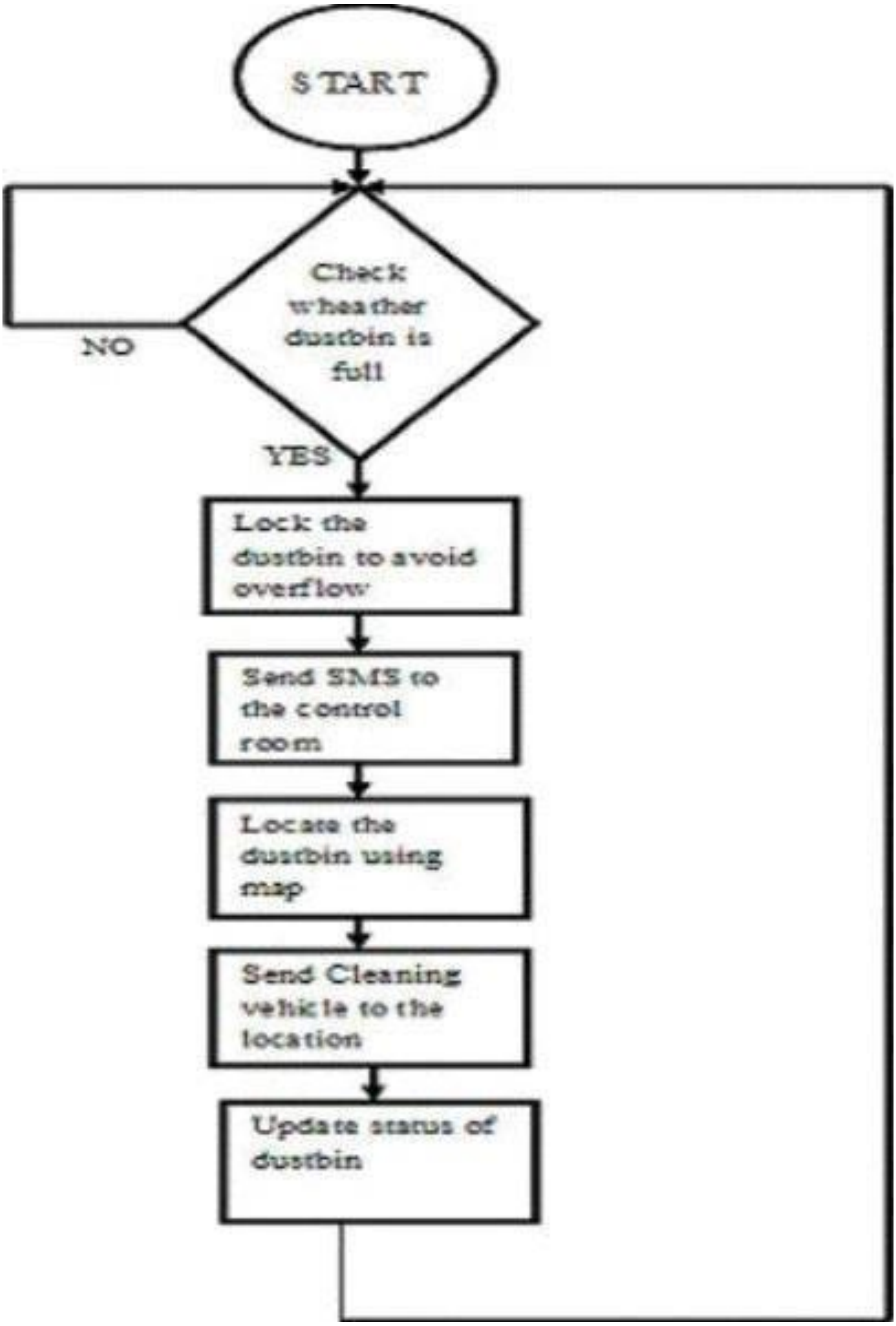


FIG:Sequence diagram

4 5.5 Flow chart

A flowchart is a picture of thc scparatc stcps of a proccss in scqucntial order. It is a gcncric tool that can be adaptcd for a wide variety of purposcs, and can bc uscd to describe various proccsscs, such as a inanufacmring proccss, an adniinistrativc or scrvicc proccss, or a projcct plan.



CHAPTER RESULTS AND DISCUSSION

5.1 RESULTS

Front thc cxisting systcin, this project advanccs in features likc maps and thc automatic lock of the lid and rcinotc acccss. Nowadays, maps play a major iinportancc in various applications likc finding a shortcst route among various others, distance, to locatc a vcliiclc likc cabs etc,. In this project, the maps arc uscd to locatc thc dustbins and helps the rcspcctivc authority to gct quick and easy to thc location. The locking systcin is used to avoid ovcrflow of trash and thc sprcading of foul smell around thc area. Evcn aftcr thc dustbin is full, pcoplc may try to dispose trash into dustbin. But thc dustbin is dcsigncd in such a way that thc lid is lockcd. This may confuse the person, literate or illiterate, that the dustbin is malfunctioning. So, to alcu thc pcrson that the garbagc is full, LED lights can bc uscd. The LED lights has to be placcd in tlic front, in a easily visiblc place. Whcn thc dustbin is full the LED light has to mm into red color. Whcn thc dustbin is not full the LED light has to turn into grccn color.

For the people in the control room to understand the status of the dustbin, a GUI is used. It can either be a website or an app. It should provide the garbage locations using maps and the status of the garbage. The application database also holds the details of the respective authorities and their details.

In addition to the features mentioned, the system can be further enhanced by incorporating real-time data analytics. By continuously monitoring the fill levels of dustbins and collecting usage patterns, the system can predict peak disposal times and suggest optimized waste collection routes. This not only ensures timely garbage pickup but also reduces fuel consumption and operational costs for waste management departments. Predictive maintenance can also be enabled through sensors, which monitor the internal condition of the dustbins, identifying potential issues such as lid malfunctions, battery levels, or sensor damage before they cause failure.

Moreover, to ensure inclusivity and broader usability, the system can support multiple languages and voice alerts to guide users when the dustbin is full or unavailable. A QR code can be added on each dustbin, which, when scanned, displays the current fill level, next expected collection time, and instructions in local languages. This promotes better public understanding and usage, thereby improving hygiene and cleanliness. Integration with municipal dashboards will also enable authorities to generate reports, track performance, and plan infrastructure improvements effectively based on data-driven insights.

Thc prototype of the idca is crcatcd for bcttcr understanding ot the projcct. Thc iinagcs of empty, half fillcd and completely filled dustbins arc below. Empty Bin (0‘ o capacity) (White LED is on HIGH):

Fig 5.1: Empty bin

Half filled Bin (50% capacity) (White and Green LED is on HIGH):

Fig 5.2: Halffilled bin.

Completely filled Bin (100% capacity) (White, Green and Red LED is on HIGH) •

Fig 5.3: Completely filled bin

When the dustbin is cinpty, thc locking system is OFF and the dustbin opcns. Whcn the dustbin is full, thc locking system is ON and thc dustbin docs not opcn.

Empty dustbin with locking system on OFF (White LED is on HIGH) •

Fig 5.4: Empty dustbin with locking system on OFF

Completely filled dustbin with locking system on ON (White, Green and Red LED is on HIGH) •

Fig 5.5: Completely filled dustbin with locking system on "ON"

When the dustbin is full, a call is sent to the control room by the GSM module. This is an indication of the coinplctcly filled dustbin. The authorities can take actions based on the call received trom the respective bins.

Indication of completely hlled bin by call using GSM module (White, Green and Red LED is on HIGH) •

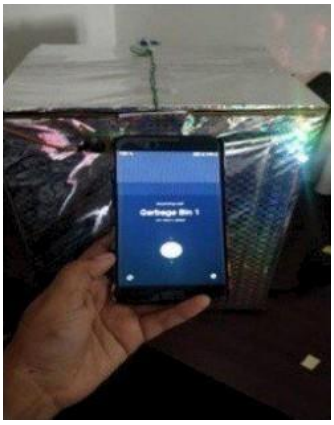


Fig 5. GSM module initiating calls

Hcrc in this project, we use the application callcd Blynk - IOT for Arduino, ESPS26f/32, Raspberry Pi. It hclps us to conncct our project in rcniotc using Blynk scrvcr. This app connects to intcmct with the hclp of Wi-Fi in the sinau phone rather than dcpcnding upon a scparatc Wi-Fi inodulc. To conncct the project to thc intcmct, download the blynk application from thc android play store. To conncct this app to our project, we need to install the necessary drivers and libraries in thc Arduino IDE. To download thc ncccssaiy drivcrs to bc ablc to usc Arduino Uno, opcn the Arduino IDE again, click on tools, boards, then boards nianagcr.

In thc search bar, scarch for " Intel curic boards". In that sclcct the vcrsion as Arduino IDE and click on install to download the drivers.

Once thc drivers arc installcd, we nccd to download thc libraries. Soinc of thc libraries that we need to run this project arc WiFi101 library, Blynk library, Ultrasonic library and so on. All thcsc libraries can bc tound in thc Arduino IDE's built-in nianagcr. To download the libraries, go to skctch, includc libraries and library inanagcr. In thc scarcli bar, scarcli for tlic acquired libraries to run the project. After installing all thc ncccssary drivers and libraries, opcn thc Blynk app and crcatc a ncw project by cntcring thc project name, dcvicc type, conncction typc and thnic.

Fig 5.7: BLYNK - Create new project.

Once crcating thc projcct, add thc ncccssary widgcts on the scrccn. In tliis Project I havc uscd 3 LEDs to know the lcvcl oí thc garbagc íillcd in the dustbin and a Map to locatc the dustbin. The LEDs arc Grccn, Orangc and Rcd indicators ivhich rcprcscnts cnipty(O‘ o), hilll fillcd(50%) and coinplctcly fillcd(lOO%) dustbin. Oncc thc capacity oí the dustbin rcaclics the thrcshold lcvcl, all the LEDs will glow.

To con nect the Arduino IDE and the Blyn k project, we need to use the Authentication token%uHieht. ümtidh @ t ÏÈÏfCta›ftdJ‹zzcaótzgnunicati¢tzc CC Mentlniuino IDE and the Blynk server.

Empty Bin (0% capacity):

Fig 5.8: BLYNK - Empty bin

Hall filled Bin (50% capacity):

Fig 5.9: BLYNK - Halffilled bin

Completely filled Bin (100% capacity):

Fig 5.10: BLYNK – Completely filled bin

To acccss the maps through remote access, thc map widgct is uscd. To get the cxact location of thc Dustbin, gct the coordinatcs from the GPRS inodulc front thc IDE and fccd it to the Blynk app to gct it on maps.

Fig 5.11: BLYNKMaps

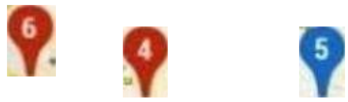


Fig 5.12: BLYNK- Shows thc'location of thc filled dustbins

5.2 COMPARISON GRAPH

READING (cm) 10	ACTUAL READING (cm) 9 15 19
40	

Tablc 5.1: Ultrasonic sensor, distance rcading vs actual readings

Ultrasonic Sensor Distance

VS Actual Distance

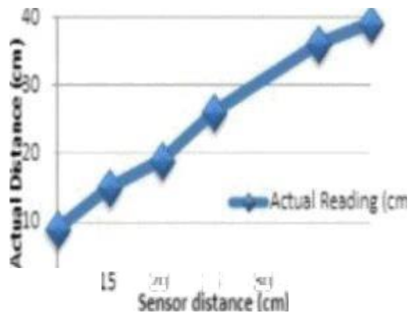


Fig 5.13: Graph of cnsor distancc vs actual distance

CONCLUSION AND FUTURE ENHANCEMENT

6.1 CONCLUSION

This project helps us to kccp the area ncat and clcan. This system also helps to monitor if thc authoritics arc doing thc allotted work in their region. This system reduces the total number ot trips ot garbagc collcction vchiclc as the systcin alcos oncc the dustbin is full and locates thc area using maps. Hence it rcduccs the ovcrall cxpcndimrc associatcd with thc garbagc collection. Since the LED lights (rcd and grecn) arc used, litcratc or thc illitcratc will bc albc to undcrstand the stams of thc dustbin. We can also help domestic animals likc dogs, cats, goats, cows etc from cating plastics. Tlicrcforc this system is uscd to crcatc a clcan cnvironmcnt with cfficicnt trash inanagcnicnt system.

6.2 FUTUREWORK

The smart dustbins can bc made niorc smartcr by cnliancing soinc inorc fcaturcs likc intcr conncction and communication among thc dustbins using IOT. Thc dustbin can also bc dcsigncd in such a way that onc large bin must havc various divisions to scgrcgatc all thc trash. Trash can be scparatcd whcn disposal as liquid waste, dccoinposablc wastc, non dcconiposablc wastc. This can also bc donc using nioismrc scnsor. By this, thc inanagcincnt of wastc will become much more casicr.

To further improve the functionality, smart dustbins can be equipped with RFID tags or barcode scanners that allow users to identify the type of waste before disposal. This ensures proper categorization and aids in automated waste sorting, which can significantly reduce manual labor at recycling centers. The data collected from these scans can also be stored in a centralized database to track waste generation patterns by region or even by individual users, promoting accountability and encouraging responsible waste disposal habits.

Additionally, integrating solar panels on the lid of the dustbin can help power the internal electronics, making the system more energy-efficient and environmentally friendly. These solarpowered bins can operate independently in remote or off-grid areas, ensuring continuous operation. Furthermore, smart alerts via SMS or app notifications can be sent to users and authorities for scheduled maintenance, collection reminders, or if unusual items or tampering are detected. These advanced features transform the smart dustbin into a fully automated, sustainable, and intelligent waste management system suitable for modern smart cities.

APPENDICES

A. SAMPLE CODE

SMART DUSTBIN.INO:

```
#includc<Scrvo.h+ #includc
<TinyGPSm.h> #include
<SofiwarcScrial.h>

#includc <BlynkSiniplcCuricBLE.li> #includc

<CurieBLE.h> #include <Ultrasonic.h> static const int RXPin = 4,
TXPin = 3; static const uint32 t PSBaud= 4800; // You should get
Auth Token in the Blynk App.
// Go to thc Projcct Scttings (nut icon).
char auth[] — " 1 v5DITTcQRy7rHDTckOoCn2DpYHYuG1

#dcfinc BLYNK PRINT Scrial
BLEPcriphcral blcPcriphcral; WidgctLED
grccn(VI);
WidgctLED orangc(V2); WidgctLED red(V3)

SoftwarcScrial siin(5, 6);


iiit timeout; String butTcr; String nuinbcr — "+919941144517"; //->
cliangc with your number Scrvo motor l; const int trig — 9; const int
ccho — 10;

const int LED1 = 1 1. const
int LED2 = 12:
const int LED3 = 13.

int duration O;
int distancc — O; TinyGPSP
lus gps;

// Thc scrial conncction to thc GPS dcvicc SotlwarcScrial ss(RXPin,
TXPin); void
sctup(; I pinModc(4,INPUT) niotorl.
attach(3); niotorl .writc( I O); pinModc(trig
OUTPUT); pinModc(ccho INPUT);
```

```
pinModc(LED1 OUTPUT);
pinModc(LED2
OUTPUT);

sim.begin(9600);


Serial.begin(9600); delay(1000);

  blcPcriplicral.sctLocalNanic("garbagc");
 blcPcriphcral.sctDcviccNainc("                    garbagc");
  blcPcriphcral.sctAppcarance(384);
Blynk.begin(blcPcriplicral, auth); blcPcriphcral.
begin();
Scrial.println("Waiting tor conncctions. ..")i

Serial.begin(115200); ss.begin(GPSBaud):

Scrial.priiitln(F("DcviccExaniplc.ino"));      Scrial.println(F("A      siinplc

dcinonstration of    TinyGPS++   with   an attachcd GPS modulc")); library  v

Scrial.print(F("Tcsting    TinyGPS++
Scrial.println(TinyGPSPlus::libraryVcrsion());
 Scrial.println(F("by      Mikal      Hart"));
 Scrial.println();
void loop()

int    x    —    digitalRcad(4);
dclay(1000); if( x==l)

 if( distancc   15 )

inotorl          write(360);
delay(3000);

else


inotorl .writc( I O);
   }    digitalWritc(trig      ,
HIGH); digitalWritc(trig •
```

```
LOW);
  duration — pulscln(ccho , HIGH);
distancc — (duration/2) / 25.5 ; if(
distancc <— 15 ) {
digitalWritc(LED1, HIGH); grccn.
on(); orange.on(); red.on();
Scrial.println(3);

else


digitalWritc(LED1, LOW);

if( distancc <— 25 )
digitalWritc(LED2, HIGH); grccn.on();
        orangc.on();
Scrial.println(2);

else


digitalWritc(LED2, LOW);

it( distancc <= 40 )

digitalWritc(LED3, HIGH);
green.on(); Scrial
println(l); } clsc
digitalWritc(LED3,
LOW); }
```

```
if( distancc <- 15 ) if
(Scrial.availablc() O)

callNuinbcr();

il (sink.available() O)
Serial.write(sim.read()):
```

```
String  rcadscrial()     tiincout

       whilc (!siin.availablc() && timeout < 12000 )
dclay(13);
  timeout*;

it (siin.availablc())     ( rctum siin.rcadString();

void callNuinbcr() siin.print (F("
          ATD"));            sini.print
(nuinbcr);      sini.print          (F("
butter — rcadscrial();
Scrial.println( buftcr); } // This skctch displays infomiation cvcry tiinc a new scntcncc is corrcctly cncodcd. whilc (ss.available() O) if (gps.cncodc(ss.read())) displaylnfo();

if (inillis() > 5000 && gps.charsProccsscd() < 10) Scrial.println(F("No
GPS dctcctcd: chcck wiring.”))i whilc(true);
void displaylnfo()

Scrial.print(F("Location:  ’t  ));  if
(gps .10cation.is Valid())

Scrial.print(gps.location.lat(),           6);
Scrial.print(F(" ,”J)i
Scrial.print(gps.location.lng(), 6);

else

Scrial.print(F("INVALID"));
```

```
Scrial.print(F(" Date/Tiinc: ")); if
(gps. date .isValid())
  {                         Scrial.print(gps.datc.month());
Scrial.print(F("/"));     Scrial.print(gps.datc.day());
Scrial.print(F("/"));
                                             }    Scrial.print(gps.datc.ycar()); clsc
                                              }   Scrial.print(F("INVALID")); Scriol.print(F("


il(gps.time.isValid())


if(gps.time.hour()          <           10)           Scrial.print(F("O"));
Scrial.print(gps.tiinc.hour());
Scrial.print(F(":")); il (gps.time.niinutc() < 10)
Scrial.print(F("O"));
Scrial.print(gps.tiinc.ininutc());
Scrial.print(F(":")); if (gps.tiinc.sccond() < 10)
Scrial.print(F("O"));
Scrial.print(gps.tiinc.sccond());   Scrial.print(F(".   "));
if(gps.tiinc.ccntisccond()  <  10)  Scrial.print(F("O"));
Serial.print(gps.time.centisecond());

else

Serial.print(F("lNVALID"));


Serial.println();
```

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